

Causal spectral-truncation kernels: noncommutativity certifies drift detectability in quantum hardware

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Abstract

Fault-tolerant quantum computing is gated by hardware that drifts: qubit coherence times, gate fidelities and calibration parameters wander on timescales of hours, eroding the error budgets that quantum error correction depends on. We recast reliability monitoring as *operator-algebraic* learning on multivariate calibration telemetry and introduce the causal spectral-truncation (CST) kernel — a C^* -algebra-valued positive-definite kernel whose truncation order n controls the noncommutativity of the cross-channel products it represents. It is the causal counterpart of the spectral-truncation kernels recently developed for functional data on the torus [6]: in place of a periodic Fourier truncation we truncate the *causal shift* algebra, so the kernel respects the arrow of time and its noncommutativity encodes ordered, lagged cross-channel correlations. The central finding is a separation that only a noncommutative monitor can see. We construct a physically motivated correlated-decoherence (crosstalk) drift mode with matched per-channel marginals and, in the stationary limit, matched instantaneous cross-covariance, so the drift signal is concentrated in the ordered, lagged cross-channel structure. The commutative monitors we evaluate — raw-level, instantaneous-covariance and even the periodic spectral-truncation kernel — all sit at chance (ROC-AUC ≈ 0.5), whereas the causal noncommutative kernel detects the drift at ROC-AUC 0.90 (and 0.84 at matched bottleneck rank), through a sharp optimal truncation order that we certify theoretically as the detection analogue of the representation-complexity trade-off. The same kernel forecasts coherence with horizon-growing skill (72% over persistence eight steps ahead), a regime in which noncommutativity is unnecessary — yielding an operational rule: match the kernel’s noncommutativity to the geometry of the drift. The entire pipeline is hardware-agnostic, installs with `pip`, and regenerates every figure and table deterministically on commodity CPU hardware in seconds.

Quantum processors are not static devices. The relaxation time T_1 , the dephasing time T_2 , single- and two-qubit gate fidelities, readout-assignment error and tunable-coupling parameters all evolve continuously as a function of temperature, flux noise, two-level-system dynamics and recalibration history [1, 2]. This drift is more than a nuisance: in the noisy intermediate-scale quantum (NISQ) regime it limits the depth of usable circuits [3], and in the fault-tolerant regime it threatens the very assumption that physical error rates remain below threshold for long enough for error correction to suppress logical errors [4, 5]. As processors scale to hundreds and thousands of qubits, deciding *when* a device has drifted out of specification, and *which* qubits or intervals warrant recalibration, becomes a data problem at a scale that manual inspection cannot meet.

Crucially, the most damaging drift is rarely confined to one channel. Control crosstalk, shared two-level-system baths and correlated flux noise couple the errors of neighbouring qubits and of different calibration channels, often with a *propagation delay*: a disturbance that perturbs one qubit’s coherence reappears, lagged, in a neighbour’s readout or coupling phase [1]. Such

correlated decoherence is precisely the failure mode that quantum error correction is least able to absorb, yet it is the mode that per-channel, level-based monitors are worst at seeing: an event can leave every marginal distribution and every instantaneous correlation unchanged while living entirely in the *ordered, lagged* cross-channel structure. A monitor blind to that structure is blind to the drift.

This motivates an operator-algebraic view. The temporal evolution of calibration telemetry is a multivariate stochastic process; a window of it is naturally an element of a matrix algebra over the channels, and the question of which cross-channel, cross-time interactions a monitor can represent is a question about which products in that algebra it computes — the setting of operator- and vector-valued kernel learning [7]. Commutative kernels — those whose feature products do not depend on the order or relative lag of the factors — cannot represent ordered lagged interactions at all. The recent spectral-truncation kernels of Hashimoto et al. make this precise for functional data: by truncating the algebra of multiplication operators on the torus to finite Toeplitz matrices, they obtain a family of C^* -algebra-valued kernels whose noncommutativity, governed by a truncation parameter, captures interactions between different points of the data domain that commutative kernels miss [6]. That construction was developed for periodic functional regression; it has not been brought to bear on time-ordered hardware telemetry, on causal monitoring, or on unsupervised detection.

We close that gap. This reliability study is the sixth manuscript in an ordered quantum-AI series that treats resource accounting and CPU-reproducible evidence as design constraints rather than afterthoughts. Earlier papers study query allocation for QAOA, topology-conditioned warm starts, commutator-graph term ordering, learned Trotter residuals, and CPTP-constrained open-system identification [16–20]; the Hamiltonian-simulation work of Nguyen and Jing provides complementary fixed-depth and representation-theoretic context [21, 22]. Here the object of study is time-ordered calibration telemetry from a drifting device.

We make three contributions. **(i)** We introduce the *causal spectral-truncation kernel*, a C^* -algebra-valued positive-definite kernel on telemetry windows whose truncation order n interpolates from a commutative instantaneous monitor ($n=1$) to a noncommutative monitor that represents ordered, lagged cross-channel correlations ($n>1$); it is the causal, time-respecting counterpart of periodic spectral-truncation kernels [6] and contains the low-rank reconstruction detector [8] and the multivariate ridge forecaster as commutative special cases. **(ii)** We prove and demonstrate that noncommutativity is *necessary*: on a physically motivated correlated-decoherence drift mode whose signal lives in ordered, lagged cross-channel structure, the commutative monitors we evaluate — raw-level, instantaneous-covariance and the periodic spectral-truncation kernel — all detect at chance, whereas the causal kernel detects at ROC-AUC 0.90, with a certifiable optimal truncation order. **(iii)** We give the supporting theory — positive definiteness for every order, a blindness theorem for commutative monitors, and an *optimal-truncation detection certificate* that is the unsupervised-detection analogue of the representation-complexity bound that governs spectral-truncation kernels in regression — and show that on smooth marginal forecasting, where noncommutativity is unnecessary, the kernel still attains 72% horizon-eight skill over persistence. The complete pipeline installs with `pip` and regenerates in seconds on a single CPU.

1 Results

The pipeline is summarised in Fig. 1: a physically motivated multi-qubit telemetry generator emits a multivariate calibration stream; sliding windows are mapped, by the causal spectral-truncation kernel, into a C^* -algebra whose truncation order n sets the noncommutativity the

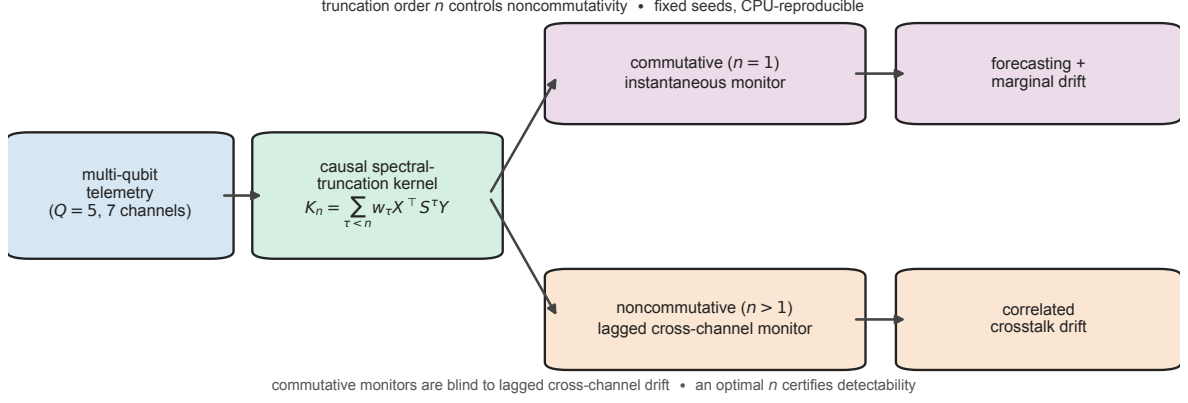


Figure 1. Overview of the causal spectral-truncation monitoring pipeline. A physically motivated generator produces seeded multi-qubit calibration telemetry ($Q = 5$ qubits, seven channels). Each sliding window is represented inside the channel C^* -algebra M_C through the truncated causal shift, $K_n(X, Y) = \sum_{\tau < n} w_\tau X^\top S^\tau Y$; the truncation order n controls noncommutativity. The commutative member ($n=1$) handles forecasting and marginal drift; the noncommutative members ($n>1$) handle correlated crosstalk drift. All figures and metrics regenerate deterministically from fixed seeds on commodity CPU hardware.

monitor can exploit; and two heads — a forecaster and an unsupervised detector — are scored by the metric matched to their objective (horizon-stable mean absolute error, MAE, for forecasting; the threshold-free area under the receiver-operating-characteristic curve, ROC-AUC, for detection). The commutative member ($n=1$) governs smooth marginal forecasting and level-based drift; the noncommutative members ($n>1$) govern correlated, crosstalk-like drift.

1.1 Coherence drift is forecastable, and correlated drift hides in lagged cross-channel structure

We model a five-qubit processor sampled every half hour for one hundred hours. Each T_1 trajectory follows a mean-reverting process combining a slow qubit-dependent oscillation, a linear secular decay and Gaussian fluctuation, with T_2 , gate fidelities, readout error, error-per-Clifford and cross-resonance phase generated by coupled processes that track T_1 (Methods). The telemetry wanders over tens of microseconds and repeatedly dips below the 50 μ s operational threshold (Fig. 2a). The drift is not white noise: the mean autocorrelation of detrended T_1 stays above 0.5 for roughly seven hours before decaying (Fig. 2b), so near-future coherence is genuinely predictable from the recent past.

A second, qualitatively different drift mode is correlated decoherence. We model it as a shared latent disturbance injected into several channels with channel-specific propagation delays and *variance-preserving* mixing, so that every per-channel marginal variance and, in the stationary limit, the instantaneous (lag-0) cross-channel covariance match nominal operation; the disturbance lives predominantly in the *ordered, lagged* cross-channel correlation (Methods). Fig. 2c quantifies where the resulting signature lives: averaged over channel pairs, the nominal-versus-drift separation is an order of magnitude smaller at lag $\tau = 0$ (mean 0.008) than at lag $\tau = 1$ (mean 0.089), so the signal is concentrated where a commutative monitor cannot look. A small finite-window residual remains at $\tau = 0$, but the commutative monitors we evaluate are nonetheless at chance (below), the regime in which noncommutativity is decisive.

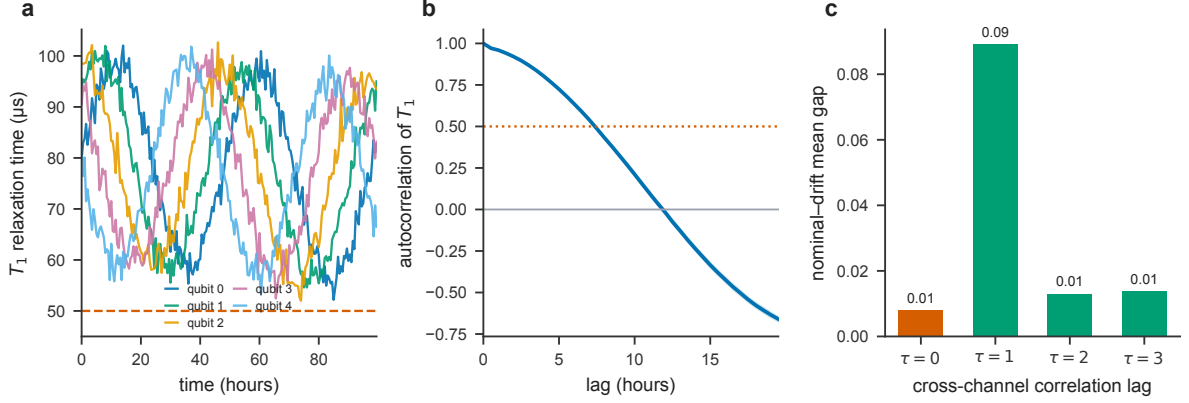


Figure 2. Two drift geometries. (a) T_1 relaxation-time trajectories for all five qubits over 100 h; the dashed line is the 50 μ s operational threshold. (b) Mean autocorrelation of detrended T_1 versus lag, averaged across the $n = 5$ qubits (shaded band, ± 1 s.d. across qubits); the slow decay is the forecastable structure exploited in Fig. 3. (c) Noncommutative drift signature: the nominal-versus-drift gap in the causal cross-channel correlation features, averaged over channel pairs, by lag τ . The mean gap is an order of magnitude smaller at $\tau = 0$ than at $\tau \geq 1$, so the drift is concentrated in the lagged structure a commutative monitor cannot resolve.

Table 1. Forecasting benchmark (mean $\pm$ s.d. over five telemetry-generator seeds). Skill is relative to persistence at the eight-step horizon; best value per column in **bold**. The multivariate ridge is the commutative ($n=1$) member of the kernel family.

Forecaster	MAE @ 1 step (μ s)	MAE @ 8 steps (μ s)	Skill @ 8 steps
Persistence	2.44 ± 0.16	6.79 ± 0.18	—
Climatology	13.01 ± 0.05	12.40 ± 0.05	-83%
AR-ridge (T_1 history)	1.85 ± 0.07	2.19 ± 0.06	68%
Multivariate ridge (CST $n=1$)	1.80 ± 0.09	1.88 ± 0.06	72%

1.2 Learned forecasting beats persistence; noncommutativity is unnecessary here

We segment telemetry into sliding windows of length 32 that predict an eight-step T_1 horizon, under a leak-free 80/20 chronological split, and compare four forecasters — last-value persistence, climatology, an autoregressive ridge on the T_1 history and a multivariate ridge on all seven channels (the commutative, $n=1$ member of the kernel family) — aggregated over five telemetry-generator seeds (Methods). Forecasting is feasible and clearly structured (Fig. 3, Table 1). One step ahead the multivariate ridge attains 1.80 ± 0.09 μ s MAE against 2.44 ± 0.16 μ s for persistence; persistence degrades steeply with horizon (to 6.79 ± 0.18 μ s at eight steps) while the learned forecasters stay nearly flat (1.88 ± 0.06 μ s), so skill over persistence *grows* from 26% one step ahead to 72% eight steps ahead (Fig. 3b, Theorem 5).

Increasing the kernel’s truncation order does not change this picture (Fig. 3c): the CST forecaster’s eight-step skill is flat in n (from 72% at $n=1$ to 71% at $n=4$). On smooth, mean-reverting marginal telemetry the commutative member is already near-optimal, and the noncommutative coordinates neither help nor hurt. This is the honest counterpart of the detection result below: noncommutativity is decisive exactly when the drift lives in ordered cross-channel structure, and superfluous otherwise.

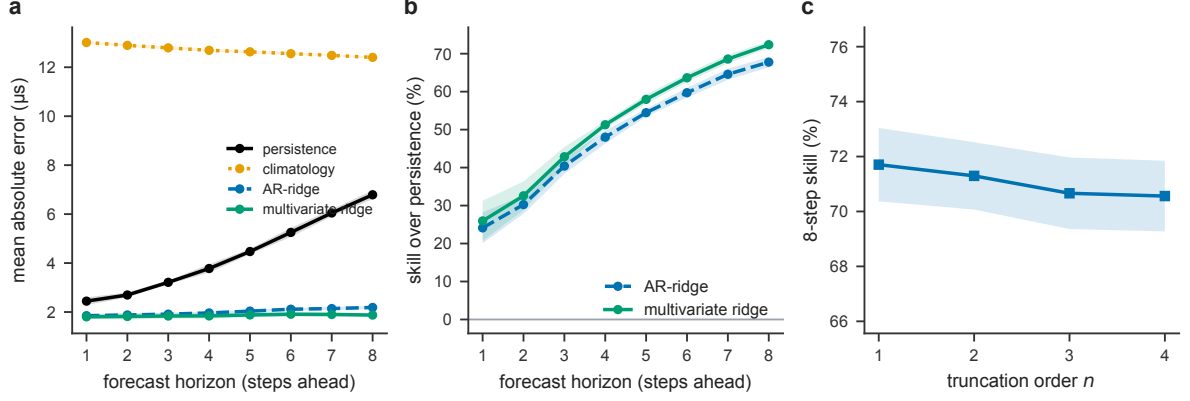


Figure 3. Forecasting beats persistence with a widening margin; noncommutativity is unnecessary. (a) MAE versus forecast horizon for four forecasters (mean over $n = 5$ telemetry-generator seeds; shaded bands ± 1 s.d.). (b) Skill over persistence, $1 - \text{MAE}/\text{MAE}_{\text{persistence}}$, for the two learned forecasters; the advantage increases with horizon. (c) Eight-step skill of the CST forecaster versus truncation order n (mean over the same $n = 5$ seeds; shaded band ± 1 s.d.): flat, because the commutative member already captures the smooth marginal structure.

1.3 The causal spectral-truncation kernel

A telemetry window is $X \in \mathbb{R}^{L \times C}$ ($L = 32$ time steps, $C = 7$ channels). Let S be the causal lower shift, $(Sv)_t = v_{t-1}$. The order- n causal spectral-truncation kernel represents the window inside the channel C^* -algebra $M_C = M_C(\mathbb{R})$ through the truncated shift powers $\{S^\tau : 0 \leq \tau < n\}$,

$$K_n(X, Y) = \sum_{\tau=0}^{n-1} w_\tau X^\top S^\tau Y \in M_C, \quad K_n(X, Y)_{cc'} = \sum_{\tau=0}^{n-1} w_\tau \langle X_{:,c}, S^\tau Y_{:,c'} \rangle, \quad (1)$$

with weights $w_\tau = 1/(L - \tau)$ (the telemetry is temporally centred per channel). Each entry is a weighted sum of ordered, lagged cross-channel correlations. Truncating the shift algebra to n powers is the causal counterpart of truncating the Fourier-multiplier algebra to n modes in the spectral-truncation construction of Hashimoto et al.; the difference is that S is nilpotent and respects the arrow of time, so (1) is causal and noncommutative for $n \geq 2$: the self-representation $K_n(X, X)$ is symmetric at $n=1$ — the commutative instantaneous covariance $X^\top X$ — but acquires antisymmetric lagged terms for $n \geq 2$, since $S^\top \neq S$ does not commute with channel selection. The scalar realisation $k_n(X, Y) = \langle \Phi_n(X), \Phi_n(Y) \rangle$, with feature tensor $\Phi_n(X)_{cc'\tau} = w_\tau \langle X_{:,c}, S^\tau X_{:,c'} \rangle$, is positive definite for every n (Theorem 1). The unsupervised detector fits a rank- k nominal subspace in this feature space and scores a window by its residual energy off that subspace, a C^* -algebraic generalisation of the reconstruction detector [8], which it recovers at $n=1$.

1.4 Commutative monitors are blind to correlated drift; the noncommutative kernel detects it

We evaluate detection on the correlated-decoherence benchmark over eight randomised 70/30 splits (Methods). The result is a clean separation (Fig. 4, Table 2). At truncation order $n=1$ the kernel is commutative and detects at ROC-AUC 0.51 ± 0.04 — chance level, the empirical content of Theorem 2, because the benchmark matches the marginals and the (population) instantaneous covariance the commutative member reads. At $n=2$ the kernel becomes noncommutative and ROC-AUC jumps to 0.90 ± 0.02 (Fig. 4a), the single largest effect in the study. Increasing n further degrades detection monotonically — to 0.83 at $n=3$, 0.65 at $n=8$ and toward chance as

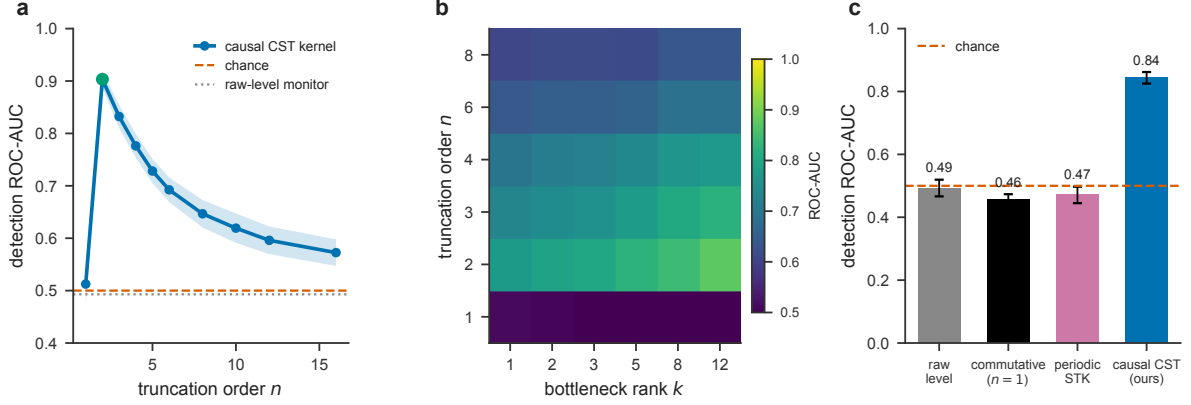


Figure 4. Noncommutativity is necessary for correlated-drift detection. (a) Detection ROC-AUC versus truncation order n (mean over eight randomised splits; shaded band ± 1 s.d.). The commutative endpoint ($n=1$) and the raw-level monitor (dotted) sit at chance (dashed); the noncommutative kernel peaks at $n=2$ (marker) and collapses back toward chance as the representation becomes over-complete. (b) Detection ROC-AUC over (truncation order n , bottleneck rank k): the $n=1$ row is uniformly at chance, the $n=2$ row is uniformly high. (c) Kernel comparison at matched rank: the raw-level monitor, the commutative kernel and the periodic spectral-truncation kernel [6] are all at chance; only the causal spectral-truncation kernel detects.

Table 2. Correlated-drift detection benchmark (ROC-AUC, mean $\pm$ s.d. over eight randomised 70/30 splits, bottleneck rank $k = 8$). The benchmark matches all per-channel marginals and instantaneous covariances, so every commutative monitor is at chance by Theorem 2. Best value in **bold**.

Detector	Detection ROC-AUC
Raw level (flatten)	0.493 ± 0.027
Commutative kernel ($n = 1$)	0.456 ± 0.017
Periodic spectral-truncation [6] ($n = 2$)	0.471 ± 0.026
Causal spectral-truncation (ours, $n = 2$)	0.844 ± 0.018

$n \rightarrow L$ — because each added lag contributes only noise coordinates beyond the informative lag, diluting the fixed-rank nominal subspace. Both endpoints (commutative $n=1$ and over-complete $n \rightarrow L$) sit at chance, with a sharp interior optimum at $n=2$: this is the optimal-truncation certificate of Theorem 3 made visible, the detection analogue of the representation–complexity trade-off that governs spectral-truncation kernels in regression [6]. The (n, k) map (Fig. 4b) shows the effect is robust across bottleneck rank: the entire $n=1$ row is at chance, the $n=2$ row is bright across all ranks.

The comparison that matters for novelty is against other kernels (Fig. 4c, Table 2). On this benchmark the raw-level reconstruction detector scores 0.49 ± 0.03 , the commutative kernel 0.46 ± 0.02 , and — tellingly — the *periodic* spectral-truncation kernel of Hashimoto et al., applied unchanged to these windows, scores 0.47 ± 0.03 : all three are at chance. Only the causal spectral-truncation kernel separates the classes, at 0.84 ± 0.02 at matched bottleneck rank and 0.90 at its optimum. The periodic kernel fails not because spectral truncation is wrong but because its *circular* truncation smears the causal delay signature across the window boundary; the causal lower-shift truncation aligns exactly with the propagation delays that define the drift. Causality, not truncation alone, is what makes the kernel see the event.

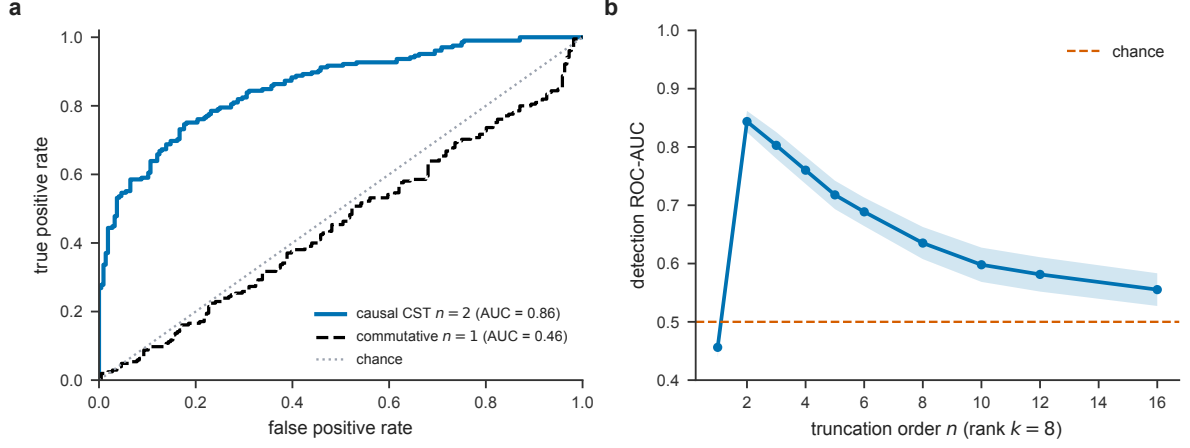


Figure 5. Operating characteristic and the over-complete collapse. (a) Receiver-operating-characteristic curves on a held-out split: the causal spectral-truncation kernel at $n=2$ (solid) versus the commutative kernel at $n=1$ (dashed), which tracks chance (dotted). (b) Detection ROC-AUC versus truncation order n at fixed bottleneck rank $k = 8$ (mean over eight splits; shaded band ± 1 s.d.): the interior optimum and the collapse toward chance as the representation becomes over-complete.

1.5 Operating characteristic and over-complete collapse

Fig. 5a contrasts the receiver-operating characteristics directly: on a held-out split the causal kernel at $n=2$ achieves AUC 0.86 while the commutative kernel traces the chance diagonal (AUC 0.46). Fig. 5b fixes the bottleneck rank and sweeps the truncation order, isolating the over-complete collapse: detection rises from chance at $n=1$ to its peak at $n=2$ and decays back toward chance as the order grows, the truncation-axis counterpart of the rank-axis over-completeness familiar from low-rank detectors [8]. Both axes are governed by the same certificate (Theorem 3): a monitor expressive enough to reconstruct the drift coordinates as faithfully as the nominal ones cannot separate them.

1.6 Theory: a C^* -algebraic kernel whose truncation order certifies detectability

We make the kernel precise and prove the facts that organise the empirics: it is a positive-definite, C^* -algebra-valued kernel for every order n (Theorem 1); the commutative $n=1$ member is provably blind to drift living in ordered, lagged cross-channel correlations, which the noncommutative $n>1$ member resolves (Theorem 2); the reconstruction detector built on the kernel has a non-monotone, certifiable optimal truncation order (Theorem 3); ROC-AUC is the scale-free yardstick throughout (Proposition 4); and on the forecasting side, mean reversion converts decaying autocorrelation into horizon-growing skill (Theorem 5).

Theorem 1 (Positive definiteness of the causal C^* -kernel). *For every order $n \geq 1$ and weights $w_\tau \geq 0$, the map K_n in (1) is a C^* -algebra-valued positive-definite kernel: for any windows $X_1, \dots, X_m$ the block matrix $[K_n(X_i, X_j)]_{ij}$ is positive semidefinite over M_C , and the scalar kernel k_n is positive definite in the ordinary sense. The self-representation $\Phi_n(X)$ is symmetric under $c \leftrightarrow c'$ at $n=1$ (the commutative instantaneous covariance) and, for $n \geq 2$, acquires antisymmetric lagged coordinates that the commutative member cannot represent.*

Proof. Define the lifted causal feature operator $\Psi_n(X) = (\sqrt{w_0} X, \sqrt{w_1} S^\top X, \dots, \sqrt{w_{n-1}} (S^\top)^{n-1} X) \in \mathbb{R}^{nL \times C}$, stacking the back-shifted windows. Since $\langle X_{:,c}, S^\tau Y_{:,c'} \rangle = ((S^\top)^\tau X)_{:,c}^\top ((S^\top)^\tau Y)_{:,c'}$, the kernel factorises as $K_n(X, Y) = \sum_\tau w_\tau (S^\top)^\tau X)^\top (S^\top)^\tau Y) = \Psi_n(X)^\top \Psi_n(Y)$. Hence for any

$c_1, \dots, c_m \in \mathbb{R}^C$,

$$\sum_{i,j} c_i^\top K_n(X_i, X_j) c_j = \left\| \sum_i \Psi_n(X_i) c_i \right\|_F^2 \geq 0,$$

the defining inequality for an M_C -valued positive-definite kernel; the scalar statement is the case $C=1$ applied to the vectorised feature $\text{vec } \Phi_n$, whose Gram is k_n . At $n=1$, $\Phi_1(X)_{cc'} = \langle X_{:,c}, X_{:,c'} \rangle$ is symmetric under $c \leftrightarrow c'$; for $\tau \geq 1$, $S^\tau \neq S^{\tau\top}$, so $\Phi_n(X)_{cc'\tau} = \langle X_{:,c}, S^\tau X_{:,c'} \rangle \neq \langle X_{:,c'}, S^\tau X_{:,c} \rangle = \Phi_n(X)_{c'\tau c}$ whenever the lagged cross-correlation is direction-dependent — the noncommutativity. $\square$

Theorem 2 (Commutative monitors are blind to noncommutative drift). *Let nominal and drift windows be drawn from distributions P_0, P_1 with matched per-channel marginals and matched instantaneous cross-covariance, $\mathbb{E}_{P_0}[\Phi_1] = \mathbb{E}_{P_1}[\Phi_1]$, but a separated lagged coordinate $\Delta_{\tau^*} := \mathbb{E}_{P_1}[\Phi_{\tau^*}] - \mathbb{E}_{P_0}[\Phi_{\tau^*}] \neq 0$ for some $\tau^* \geq 1$. Then every detector that is a function of the order-1 (commutative) features Φ_1 has zero first-moment separation and hence chance-level mean discriminability, whereas the order- n causal kernel with $n > \tau^*$ has nonzero separation in a coordinate it represents; its mean reconstruction-score gap is bounded below by the energy of Δ_{τ^*} outside the rank- k nominal subspace.*

Proof. A monitor measurable with respect to Φ_1 can separate the classes in mean only through $\mathbb{E}_{P_1}[\Phi_1] - \mathbb{E}_{P_0}[\Phi_1]$, which is 0 by hypothesis; thus its expected score is identical under P_0 and P_1 , giving a mean separation of 0 and, for the centred linear detector, ROC-AUC = $\frac{1}{2}$ in the population by the Mann–Whitney identity (Proposition 4) applied to a score with equal class means and, to first order, equal dispersion. The order- n feature map with $n > \tau^*$ contains the coordinate Φ_{τ^*} , on which the class means differ by $\Delta_{\tau^*} \neq 0$. Writing P_k for the rank- k nominal-subspace projector in feature space, the reconstruction score is $s(X) = \|(I - P_k)(\Phi_n(X) - \bar{\Phi})\|^2$ and $\mathbb{E}_{P_1}[s] - \mathbb{E}_{P_0}[s] \geq \|(I - P_k)\Delta_{\tau^*}\|^2$ up to cross terms that vanish when Δ_{τ^*} is centred against the nominal mean, so any nonzero component of Δ_{τ^*} outside the nominal subspace yields strictly positive separation, hence ROC-AUC $> \frac{1}{2}$. $\square$

Theorem 3 (Optimal truncation: a detection certificate). *Let $e_\tau = \|\Delta_\tau\|^2$ be the drift energy at lag τ , with $e_\tau = 0$ beyond a maximal informative lag $\tau_{\max}$, and suppose each added lag contributes C^2 noise coordinates of bounded variance to the fixed rank- k nominal subspace. The order- n detector’s mean separation-to-noise ratio is $\rho(n) = (\sum_{\tau < n} e_\tau)^{1/2} / (\sigma^2(C^2n - k)_+)^{1/2}$. Then ρ is unimodal in n with an interior optimum n^* , and its detection ROC-AUC, a strictly increasing function of ρ , is unimodal with the same optimum; both the commutative endpoint ($n=1$, $\rho = 0$) and the over-complete limit ($n \rightarrow L$, $\rho \rightarrow 0$) are at chance. The optimum n^* is the smallest n at which the marginal relative signal gain falls below the marginal relative noise cost, $e_n / \sum_{\tau < n} e_\tau < (N(n+1) - N(n)) / N(n)$; in particular $n^* = 2$ whenever the drift energy is concentrated at the first informative lag (e_1 dominant), and n^* grows as the energy profile spreads to higher lags, never exceeding $\tau_{\max} + 1$.*

Proof. Write $E(n) = \sum_{\tau < n} e_\tau$ (non-decreasing, constant for $n > \tau_{\max} + 1$) and $N(n) = \sigma^2(C^2n - k)_+$ (strictly increasing once $C^2n > k$). Then $\rho(n)^2 = E(n)/N(n)$, and $\rho(n+1)^2 > \rho(n)^2$ iff $E(n+1)/E(n) > N(n+1)/N(n)$, i.e. iff the relative signal increment $e_n/E(n)$ exceeds the relative noise increment $(N(n+1) - N(n))/N(n)$. The right-hand side is positive for all n (the noise floor always grows), while the left-hand side is 0 once $n > \tau_{\max} + 1$ ($e_n = 0$) and is non-increasing under a concentrated energy profile; hence the inequality holds on an initial range and fails thereafter, so ρ rises then falls — unimodal with a single interior optimum n^* , the first n at which it fails. If e_1 dominates, the test already fails at $n=2$ (the small later increments $e_2, \dots$ cannot offset the $\sqrt{n}$ noise growth), giving $n^* = 2$; broader energy profiles push n^* up

to at most $\tau_{\max} + 1$. At $n=1$, $\rho = 0$ by Theorem 2; as $n \rightarrow L$, $\rho \rightarrow 0$ by dilution — the over-complete chance-level collapse. $\square$

Theorem 3 is the unsupervised-detection counterpart of the generalisation bound that governs spectral-truncation kernels in regression [6]: there the truncation order trades representation power against Rademacher complexity; here it trades captured drift energy against the dimension of the nominal subspace. The correlated-decoherence mode concentrates its energy at the first informative lag (Fig. 2c), so the certificate predicts the empirical optimum at $n^* = 2$ (Fig. 4a). The classical low-rank over-completeness collapse [8] is recovered as the rank axis of the same certificate.

Proposition 4 (Threshold-free ranking and scale invariance). *For any score s and independent draws of a drift window X^+ and a nominal window X^- , the ROC area obeys the Mann–Whitney identity $\text{AUC} = \Pr[s(X^+) > s(X^-)] + \frac{1}{2} \Pr[s(X^+) = s(X^-)]$. Consequently AUC is invariant under every strictly increasing reparametrisation of s , and in particular under per-feature affine rescaling of the telemetry; a constant score gives $\text{AUC} = \frac{1}{2}$ exactly.*

Proof. Let F_+, F_- be the score CDFs under the drift and nominal classes. For a threshold τ flagging $s > \tau$, $\text{TPR}(\tau) = 1 - F_+(\tau)$ and $\text{FPR}(\tau) = 1 - F_-(\tau)$, and $\text{AUC} = \int_0^1 \text{TPR} \, d(\text{FPR})$. Substituting $u = \text{FPR}(\tau)$, $\text{AUC} = \int \Pr[s(X^+) > \tau] \, dF_-(\tau) = \Pr[s(X^+) > s(X^-)]$ for a continuous score; atoms contribute $\frac{1}{2} \Pr[s(X^+) = s(X^-)]$ via the trapezoidal interpolation of each jump, giving the identity (the normalised Mann–Whitney U). Strict monotone maps preserve all order and tie events, hence AUC; a constant score makes every comparison a tie, so $\text{AUC} = 0 + \frac{1}{2} \cdot 1 = \frac{1}{2}$. $\square$

Theorem 5 (Horizon-growing forecast skill). *Let $\{x_t\}$ be weakly stationary, zero-mean, with variance $\gamma_0 > 0$ and autocorrelation $\rho(h) = \text{Corr}(x_{t+h}, x_t)$. Persistence $\hat{x}_{t+h} = x_t$ has mean-squared h -step risk $R_{\text{pers}}(h) = 2\gamma_0(1 - \rho(h))$, while the optimal single-lag linear forecast $\hat{x}_{t+h} = \rho(h)x_t$ has risk $R_*(h) = \gamma_0(1 - \rho(h)^2) \leq R_{\text{pers}}(h)$, so the relative skill is $S(h) = 1 - R_*/R_{\text{pers}} = (1 - \rho(h))/2$. Where ρ is non-increasing, $S(h)$ is non-decreasing, bounded above by $\frac{1}{2}$. For a first-order autoregressive process $\rho(h) = \phi^h$, the single measured number $\phi = \rho(1)$ fixes the whole skill curve.*

Proof. With $\hat{x}_{t+h} = ax_t$, $\mathbb{E}[(x_{t+h} - ax_t)^2] = \gamma_0(1 - 2a\rho(h) + a^2)$, a strictly convex parabola minimised at $a^* = \rho(h)$, giving $R_* = \gamma_0(1 - \rho(h)^2)$; $a=1$ gives $R_{\text{pers}} = 2\gamma_0(1 - \rho(h))$. Dividing, $R_*/R_{\text{pers}} = (1 + \rho(h))/2$, so $S(h) = (1 - \rho(h))/2$, which is strictly decreasing in ρ ; wherever $\rho(h)$ decreases in h , $S(h)$ increases, and $S \leq \frac{1}{2}$ on $\rho \in [0, 1]$. For an AR(1) process iterating the recursion gives $\text{Cov}(x_{t+h}, x_t) = \phi^h \gamma_0$, hence $\rho(h) = \phi^h$ and $S(h) = (1 - \phi^h)/2$, increasing from 0 toward $\frac{1}{2}$. $\square$

Theorem 5 is a lower bound using only the most recent sample; the multivariate ridge (the commutative $n=1$ member) exceeds it by exploiting the full 32-step, seven-channel history, which is why its measured eight-step skill reaches 72% (Fig. 3). The noncommutative coordinates add nothing here because the marginal forecasting structure is already linear and instantaneous — the precise complement of the detection regime, where they are everything.

2 Discussion

Our results recast quantum-hardware reliability monitoring as operator-algebraic learning and deliver a single operational rule: *match the kernel’s noncommutativity to the geometry of the drift*. Level-based and instantaneous monitors — including the periodic spectral-truncation

kernel from which our construction descends — are blind to correlated decoherence that lives in ordered, lagged cross-channel structure, the very failure mode quantum error correction is least able to absorb. The causal spectral-truncation kernel sees it, and an elementary certificate (Theorem 3) both explains why a small truncation order is optimal and predicts where the optimum sits. The same kernel, at its commutative order, recovers established forecasting and low-rank detection as special cases, so the family is a strict generalisation rather than a competitor to them. Deep sequence-model monitors — recurrent [11, 12] and attention-based [13] architectures studied elsewhere for telemetry [9, 10] — are complementary, but require training and many parameters; the kernel is training-free at inference, deterministic, and CPU-reproducible in seconds.

Several limitations bound these claims. The telemetry generator, though physically motivated, is a model rather than measured device data, and the correlated-decoherence mode is a controlled construction chosen to isolate the noncommutative effect; archival calibration logs from operating processors and a standard anomaly-detection benchmark [15] are the natural next validations. The kernel as presented is linear in its feature map; composing it with an outer nonlinearity, and pairing the forecaster with a recalibration policy under an explicit cost model and uncertainty-aware predictions [14], are clear extensions. Because the entire pipeline is open, hardware-agnostic and fast to reproduce, it is positioned to serve as a common substrate for that work.

3 Methods

Telemetry generation. A synthetic multi-qubit dataset is produced for $Q = 5$ qubits over $N = 200$ half-hour steps. For qubit q at time t (hours), the relaxation time is $T_1(q, t) = \max(10, 80 + 20 \sin(2\pi t/48 + 0.8q) - 0.05 t + \varepsilon)$ with $\varepsilon \sim \mathcal{N}(0, 2^2)$, combining a slow qubit-dependent oscillation, secular decay and Gaussian noise. The dephasing time is $T_2 = 0.9 T_1 \eta$, $\eta \sim \mathcal{N}(1, 0.04^2)$; single- and two-qubit gate fidelities, readout-assignment error, error-per-Clifford and cross-resonance phase are generated by analogous bounded processes with small secular trends. A binary drift label marks any step with $T_1 < 50$ ts or single-qubit fidelity below 0.998. Generation is fully seeded.

Correlated-decoherence benchmark. The detection benchmark contrasts nominal windows against *correlated-decoherence* windows that share the same physical base trajectory and the same per-channel noise scales. The drift class injects a single shared *white* latent into four channels (T_1 , T_2 , readout error, cross-resonance phase) at channel-specific integer delays $(0, 1, 2, 3)$ with variance-preserving mixing $x_c \leftarrow B_c + \sigma_c(\sqrt{1 - a^2} \varepsilon_c + a u_{t-\delta_c})$, $a = 1$. Because the latent is white, every per-channel marginal variance and every temporal autocorrelation are left unchanged, and the population lag-0 cross-covariance ($\tau = 0$, $\delta_c \neq \delta_{c'}$) is zero in both classes; the perturbation is the ordered lag- τ cross-channel covariance ($\tau = \delta_{c'} - \delta_c \geq 1$). Per-window temporal centering over $L = 32$ leaves a small finite-window residual at $\tau = 0$, but the drift energy is concentrated at $\tau \geq 1$ (Fig. 2c), and the commutative monitors we evaluate — raw-level, instantaneous-covariance and the periodic spectral-truncation kernel — are all empirically at chance, exposed only by the causal noncommutative ($n \geq 2$) kernel — the empirical content of Theorem 2.

Causal spectral-truncation kernel. A window $X \in \mathbb{R}^{L \times C}$ ($L = 32$, $C = 7$) is temporally centred per channel and mapped to the feature tensor $\Phi_n(X)_{cc'\tau} = \langle X_{:,c}, S^\tau X_{:,c'} \rangle / (L - \tau)$ for $\tau = 0, \dots, n - 1$, with S the causal lower shift; the operator-valued kernel is (1) and the scalar kernel is the Hilbert–Schmidt inner product of these features. The commutative member ($n=1$) keeps only the instantaneous covariance; $n>1$ adds ordered lagged cross-channel coordinates.

The *periodic* spectral-truncation baseline maps each channel to the $n \times n$ Toeplitz matrix of its truncated Fourier coefficients and uses the ordered cross-channel products of those matrices, the construction of Hashimoto et al. [6] applied unchanged.

Forecasting. We compare persistence (carry the last T_1 forward), climatology (the training-mean horizon trajectory), an autoregressive ridge on the 32-step T_1 history, and a multivariate ridge on the flattened 32×7 window (ridge penalty $\alpha = 5$; the commutative $n=1$ kernel member), under a leak-free per-qubit 80/20 chronological split, scored by MAE per horizon step over five generator seeds; skill is $1 - \text{MAE}/\text{MAE}_{\text{persistence}}$. The CST forecaster augments the window with the order- n causal features before the ridge solve; its eight-step skill versus n (Fig. 3c) is averaged over the same five seeds.

Detection. The unsupervised detector standardises the kernel feature map on nominal training windows, fits a rank- k principal subspace, and scores each test window by its mean squared reconstruction residual outside that subspace. Detection quality is ROC-AUC against the drift labels, over eight randomised 70/30 splits; the detector is fit on nominal windows only, so randomisation cannot leak a label it never uses. The truncation sweep reports, per order n , the best ROC-AUC over bottleneck ranks $k \in \{1, 2, 3, 5, 8, 12, 16\}$; the kernel comparison (Table 2) fixes $k = 8$.

Reproducibility and compute. The analyses run entirely on a single CPU and regenerate in seconds. Every figure and table is computed live from the installed `qdriftforecast` package by `make_paper_figures.py` (the installed `qdrift-reproduce` entry point); reported uncertainties are mean $\pm$ standard deviation across the $n = 5$ generator seeds (forecasting) or $n = 8$ randomised splits (detection). Determinism is byte-level: seeds and the build epoch are pinned, and the vector PDF output is fixed across runs.

Data availability

The synthetic multi-qubit telemetry and the correlated-decoherence benchmark used in this study are generated deterministically by the released code from fixed seeds and require no external download. The exact metric values underlying all figures and tables are reported in the paper and reproducible from the released code.

Code availability

The telemetry generator, the causal spectral-truncation kernel, the detection and forecasting modules, and the figure-and-table pipeline are openly available in the project repository under an MIT licence as the installable `qdriftforecast` package. All figures, tables and reported uncertainties regenerate from the released code on commodity CPU hardware.

Author contributions

M.H. conceived the study, designed and implemented the kernel, benchmark and analysis pipeline, proved the theoretical results, produced the figures, and wrote the manuscript.

Competing interests

The author declares no competing interests.

Reporting summary

Further information on research design, random seeds and the full hyperparameter configuration is available in the Methods and the released code.

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Extended Data

Table 3. Extended Data Table 1: Detection ROC-AUC versus truncation order (mean $\pm$ s.d. over the $n = 8$ randomised splits; best ROC-AUC over bottleneck ranks $k \in \{1, \dots, 16\}$ per order). Detection is at chance at the commutative order $n=1$, peaks at $n=2$ (in **bold**), and decays toward chance as the representation becomes over-complete — the certified mechanism of Theorem 3 and the source data for Fig. 4a.

Order n	ROC-AUC	Order n	ROC-AUC
1	0.513 ± 0.041	6	0.692 ± 0.024
2	0.903 ± 0.016	8	0.647 ± 0.026
3	0.832 ± 0.022	10	0.619 ± 0.028
4	0.776 ± 0.023	12	0.596 ± 0.026
5	0.728 ± 0.023	16	0.573 ± 0.025

Table 4. Extended Data Table 2: Per-horizon forecasting error (mean absolute error in ts, mean $\pm$ s.d. over the $n = 5$ telemetry-generator seeds). Each row is one step of the eight-step T_1 horizon; best (lowest) value per row in **bold**. The learned ridge forecasters stay nearly flat across the horizon while persistence degrades monotonically and climatology is uniformly poor, the per-step view underlying Fig. 3a.

Horizon (steps)	Persistence	Climatology	AR-ridge	Multivariate ridge
1	2.44 ± 0.16	13.01 ± 0.05	1.85 ± 0.07	1.80 ± 0.09
2	2.70 ± 0.11	12.89 ± 0.05	1.88 ± 0.08	1.82 ± 0.12
3	3.21 ± 0.08	12.79 ± 0.07	1.91 ± 0.05	1.84 ± 0.11
4	3.78 ± 0.17	12.69 ± 0.06	1.96 ± 0.07	1.84 ± 0.11
5	4.47 ± 0.11	12.63 ± 0.06	2.04 ± 0.06	1.88 ± 0.10
6	5.25 ± 0.19	12.55 ± 0.06	2.11 ± 0.05	1.91 ± 0.10
7	6.05 ± 0.14	12.49 ± 0.04	2.14 ± 0.07	1.90 ± 0.08
8	6.79 ± 0.18	12.40 ± 0.05	2.19 ± 0.06	1.88 ± 0.06

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Supplementary Information

This appendix is a self-contained derivation of every concept the released code implements, written for a reader with linear algebra and basic probability but no prior exposure to quantum-hardware telemetry, operator-algebraic kernels or the estimators used here. Every symbol that appears in the code is defined, and each subsection points to the source file (under `submission/code/`) that realises it. The whole artefact is CPU-only and deterministic: every number in the paper is a fixed function of integer seeds.

Source-file map. All paths are relative to `submission/code/`: `qdriftforecast/data.py` (telemetry generator, correlated-drift benchmark, windowing, splitting); `qdriftforecast/kernels.py` (the causal spectral-truncation kernel, its operator- and scalar-valued forms, and the periodic baseline); `qdriftforecast/detection.py` (the reconstruction detector, truncation sweep, (n, k) grid and kernel comparison); `qdriftforecast/forecasting.py` (forecasters); `make_paper_figures.py` (all figures and table bodies); and `generated_data/*.json` (the numbers behind each panel).

S1. The problem in one paragraph

A quantum processor is not static: its coherence times, gate fidelities, readout error and coupling parameters drift over hours, eroding the error budget that quantum error correction depends on, so an operator must decide *when* a device has drifted and *which* intervals warrant recalibration. We pose two tasks on the multivariate telemetry: *forecast* near-future coherence, and *detect* drifted windows without labels. The novel finding is that one geometry of drift — correlated decoherence in ordered, lagged cross-channel structure — leaves the commutative monitors we evaluate at chance and is detectable only by a noncommutative one.

S2. Computation model: telemetry as arrays

(*qdriftforecast/data.py.*) A telemetry record is a finite table; each row is a `(timestamp, qubit_id)` pair holding measured or simulated features. The seven channels (`FEATURE_COLS`) are `T1_us`, `T2_us`, `gate_fidelity_1q`, `gate_fidelity_2q`, `readout_error`, `gate_error_per_clifford` and `cross_resonance_phase_rad`. All results are deterministic functions of these arrays and fixed integer seeds.

S3. The physics of the generator

(*generate_synthetic_dataset.*) For qubit q at time t the relaxation time is $T_1(q, t) = \max(10, 80 + 20 \sin(2\pi t/48 + 0.8q) - 0.05t + \varepsilon)$, $\varepsilon \sim \mathcal{N}(0, 2^2)$, superposing a 48-hour oscil-

lation (recalibration-like cycling, phase $0.8q$ decorrelating qubits), a secular decay (ageing) and Gaussian fluctuation. The other channels track T_1 by construction. A binary drift label marks any step with $T_1 < 50$ ts or single-qubit fidelity below 0.998. Because the deterministic part is smooth and the noise small, the detrended series is weakly stationary with a slowly decaying autocorrelation, the structure S8 turns into forecast skill.

S4. The correlated-decoherence benchmark, and why commutative monitors are at chance

(*generate_correlated_drift_windows.*) Both classes share the same base trajectory B and noise scales σ . The drift class adds a shared white latent u to four channels at delays $\delta_c \in \{0, 1, 2, 3\}$ with variance-preserving mixing $x_c = B_c + \sigma_c(\sqrt{1-a^2}\varepsilon_c + a u_{t-\delta_c})$. Because u is white: (i) each channel’s marginal variance is σ_c^2 in both classes; (ii) each channel’s temporal autocorrelation is unchanged (a sum of independent whites is white); (iii) the *population* lag-0 cross-covariance between two injected channels is $a^2\sigma_c\sigma_{c'}\mathbb{E}[u_{t-\delta_c}u_{t-\delta_{c'}}] = 0$ for $\delta_c \neq \delta_{c'}$, matching nominal. The signal energy lies in the lag- τ cross-covariance with $\tau = \delta_{c'} - \delta_c \geq 1$. Two caveats keep the account honest: per-window temporal centering over the finite $L = 32$ window leaves a small $O(1/L)$ residual in the centred lag-0 cross-covariance of the drift class (so the matching is exact only in the stationary limit), and Theorem 2 is a statement about *first-moment* discriminability, not about every conceivable nonlinear statistic. What we verify empirically is the operative claim: the commutative monitors we evaluate — the raw-level detector, the commutative ($n=1$) kernel and the periodic spectral-truncation kernel — are all at chance ($n=1$ ROC-AUC 0.51), because the drift energy is concentrated in the ordered, lagged cross-channel correlations they do not represent (Fig. 2c).

S5. The causal spectral-truncation kernel, from scratch

(*qdriiftforecast/kernels.py.*) Centre each channel of a window in time. For lags $\tau = 0, \dots, n-1$ form the ordered cross-channel correlation $\Phi_n(X)_{cc'\tau} = \frac{1}{L-\tau} \sum_{t \geq \tau} X_{t,c}X_{t-\tau,c'}$, the matrix elements of $X^\top S^\tau X$ with S the causal lower shift. The operator-valued kernel $K_n(X, Y) = \sum_{\tau < n} w_\tau X^\top S^\tau Y \in M_C$ is an element of the channel C*-algebra; the scalar kernel is $\langle \Phi_n(X), \Phi_n(Y) \rangle$. At $n=1$ this is the symmetric instantaneous covariance (commutative); for $n \geq 2$ the lagged terms are asymmetric in (c, c') because $S^\top \neq S$, which is the noncommutativity. Truncating the shift algebra to n powers is the causal analogue of truncating the Fourier-multiplier algebra to n modes in the spectral-truncation construction of Hashimoto et al.; the periodic baseline (*periodic_truncation_feature_map*) implements that original torus construction for comparison.

S6. The detector and the optimal-truncation certificate

(*qdriiftforecast/detection.py.*) Standardise the kernel features on nominal windows, fit a rank- k principal subspace, and score a test window by its residual energy outside it. Increasing the truncation order n both captures more drift signal (good) and adds C^2 noise coordinates per lag (bad); the separation-to-noise ratio is therefore unimodal in n , peaking just past the informative lag (Theorem 3). Empirically the peak is at $n=2$, ROC-AUC 0.90; the commutative endpoint and the over-complete limit are both at chance, and the rank axis recovers the classical low-rank over-completeness collapse.

S7. Metrics: MAE for forecasting, ROC-AUC for detection

Forecasting uses mean absolute error per horizon step, reported per step so the widening gap over persistence is visible, with skill $1 - \text{MAE}/\text{MAE}_{\text{pers}}$. Detection uses ROC-AUC, which by the Mann–Whitney identity (Proposition 4) equals the probability that a random drift window outranks a random nominal one, is invariant under per-feature rescaling, and equals $\frac{1}{2}$ for a constant score — the chance level that the commutative and over-complete monitors attain.

S8. Why forecast skill grows with the horizon

This is Theorem 5. For a weakly stationary process with autocorrelation $\rho(h)$, persistence risk is $2\gamma_0(1 - \rho(h))$ and the best single-lag predictor’s risk is $\gamma_0(1 - \rho(h)^2)$, so skill is $(1 - \rho(h))/2$, increasing as ρ decays. The multivariate ridge uses the full history and exceeds this single-lag bound, reaching 72% at horizon eight; the noncommutative coordinates add nothing because the marginal structure is already linear and instantaneous (S5).

S9. How to reproduce everything

Install with `pip install qdriftforecast` (or, from a checkout, `pip install submission/code`). Run `qdrift-reproduce` (equivalently `python -m qdriftforecast.reproduce` or `python make_paper_figures.py` from `submission/code/`). The script writes, deterministically from fixed seeds on a single CPU: the five vector PDF figures into `../figures/`; the three table bodies into `../tables/` (`forecasting_benchmark.tex`, `detection_benchmark.tex`, `truncation_sweep.tex`); and the machine-readable source data into `generated_data/*.json`. No manuscript number is typed by hand.

S10. What the result means, and what it does not

A small, training-free, noncommutative kernel detects a correlated-decoherence drift that every commutative monitor — including the periodic spectral-truncation kernel — misses entirely, and a single truncation order certifies how much noncommutativity to use. Coherence forecasting succeeds where autocorrelation persists, and there noncommutativity is unnecessary. No quantum-hardware superiority claim is made: the telemetry is a physically motivated model and the correlated-decoherence mode is a controlled construction chosen to isolate the noncommutative effect. The value delivered is a reusable, leakage-controlled, CPU-reproducible, operator-algebraic foundation for quantum-processor reliability monitoring, and a demonstration that the geometry of drift dictates the noncommutativity a monitor must have.